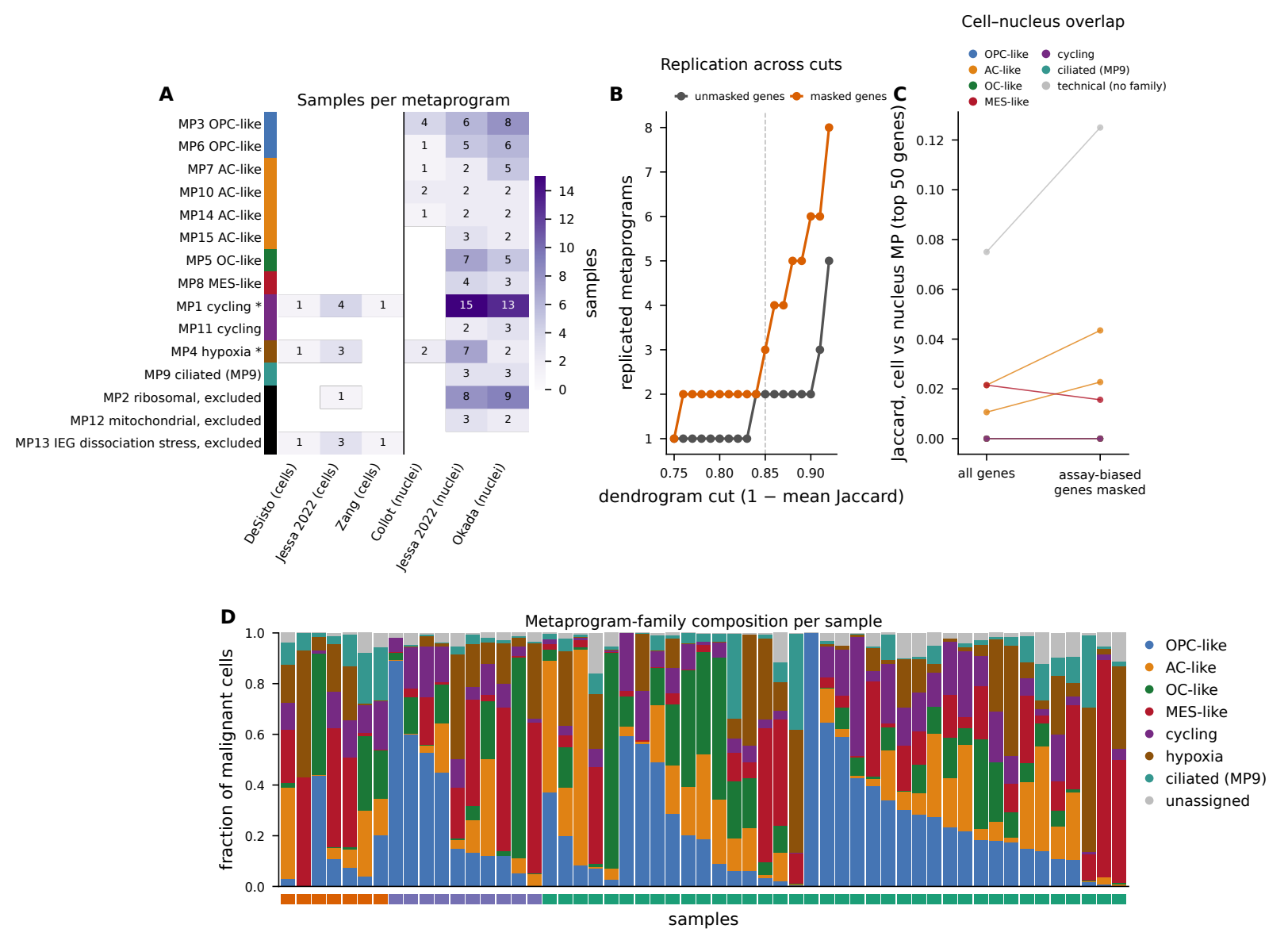
A
Samples per metaprogram
MP3 OPC-like
MP6 OPC-like
MP7 AC-like
MP10 AC-like
MP14 AC-like
MP15 AC-like
MP5 OC-like
MP8 MES-like
MP1 cycling *
MP11 cycling
MP4 hypoxia *
MP9 ciliated (MP9)
MP2 ribosomal, excluded
MP12 mitochondrial, excluded
MP13 IEG dissociation stress, excluded
DeSisto (cells)
Jessa 2022 (cells)
Zang (cells)
Collot (nuclei)
Jessa 2022 (nuclei)
Okada (nuclei)
samples
B
Replication across cuts
unmasked genes
masked genes
replicated metaprograms
dendrogram cut (1 − mean Jaccard)
C
Cell–nucleus overlap
OPC-like
AC-like
OC-like
MES-like
cycling
ciliated (MP9)
technical (no family)
Jaccard, cell vs nucleus MP (top 50 genes)
all genes
assay-biased genes masked
D
Metaprogram-family composition per sample
fraction of malignant cells
samples
OPC-like
AC-like
OC-like
MES-like
cycling
hypoxia
ciliated (MP9)
unassigned